

DUAL-FACTOR BIOMETRIC & CRYPTOGRAPHIC ATTENDANCE MANAGEMENT SYSTEM

An Autonomous Attendance Framework Integrating OpenCV LBPH Facial Biometrics and Dynamic QR Code Validation

PROJECT REPORT

Architecture: Python Flask • OpenCV LBPH • MySQL Relational Engine • Vanilla HTML5/CSS3

Domain: Computer Vision, Biometrics & Cryptographic Security

Deployment: 100% On-Premise • Touchless • Zero External Subscriptions

ABSTRACT

In contemporary educational institutions and corporate enterprises, attendance tracking is fundamental to accountability, performance evaluation, and organizational discipline. Traditional attendance auditing mechanisms—such as handwritten roll calls, magnetic stripe identification badges, radio frequency identification (RFID) tags, and optical fingerprint scanners—are increasingly susceptible to vulnerabilities. These include "proxy attendance" (buddy punching), credential theft, hygiene risks from physical contact, hardware wear, and labor-intensive manual administrative workflows.

To resolve these limitations, this project presents the design, architectural engineering, and practical implementation of a **Dual-Factor Biometric and Cryptographic Attendance Management System**. The system synthesizes two decoupled security modalities into an automated verification terminal:

- **Cryptographic Token Verification:** A physical or dynamic Quick Response (QR) code containing an encrypted, high-entropy token assigned to the registered individual.
- **Biometric Facial Recognition:** An automated computer vision pipeline combining Haar Feature-based Cascade Classifiers for frontal face detection and Local Binary Patterns Histograms (LBPH) for illumination-invariant biometric texture classification.

Built on an asynchronous Python Flask web engine and a high-performance MySQL transactional relational database, the system operates completely on-premise without reliance on paid cloud computer vision APIs, third-party biometrics vendors, or external SaaS infrastructure. The user interface employs modern HTML5, CSS3 cybernetic glassmorphism, and client-side JavaScript powered by WebRTC video capture and in-browser canvas decoding via jsQR.

Furthermore, the platform integrates automated Simple Mail Transfer Protocol (SMTP) notifications for credential distribution and scheduled absentee cutoff notifications, alongside administrative analytics and dynamic CSV reporting. The proposed architecture eliminates proxy marking, achieves sub-second verification latency, and provides a scalable, touchless, zero-maintenance operational model.

Key System Highlights

Dual-Factor Security: Token possession + Facial biometric verification.

High Performance: Sub-500ms end-to-end verification latency.

Privacy & Zero Cost: 100% on-premise local execution with no third-party APIs.

Automated Governance: Atomic daily duplicate prevention and cutoff absentee alerts.

TABLE OF CONTENTS

SECTION / TOPIC	REF
ABSTRACT	ii
1. INTRODUCTION	1
1.1 SYSTEM SPECIFICATION	1
1.2 HARDWARE CONFIGURATION	2
1.3 SOFTWARE SPECIFICATION	2
2. SYSTEM STUDY	3
2.1 EXISTING SYSTEM	3
2.1.1 DRAWBACKS	3
2.2 PROPOSED SYSTEM	4
2.2.1 FEATURES	4
3. SYSTEM DESIGN AND DEVELOPMENT	6
3.1 INPUT DESIGN	6
3.2 OUTPUT DESIGN	6
3.3 CODE DESIGN	7
3.4 DATABASE DESIGN	8
3.5 SYSTEM DEVELOPMENT	10
3.5.1 DESCRIPTION OF MODULES	10
4. TESTING AND IMPLEMENTATION	12
4.1 TESTING	12
4.1.1 FUNCTIONAL TESTING (TEST MATRIX)	12
4.1.2 IMPLEMENTATION	13
5. CONCLUSION	14

SECTION / TOPIC	REF
6. BIBLIOGRAPHY	15

1. INTRODUCTION

Attendance tracking forms the operational backbone of universities, colleges, manufacturing units, and corporate offices. Accurate attendance records influence academic eligibility, grading, payroll computation, compliance auditing, and physical site security. Over the past several decades, attendance recording has migrated from manual pen-and-paper ledgers to automated computerized solutions.

However, mainstream automated methodologies continue to suffer from fundamental design weaknesses:

- **Possession-based tokens** (such as RFID cards and barcodes) confirm only the *possession* of an object, failing to verify the physical presence of the authenticated human being.
- **Single-factor biometric solutions** (such as fingerprint readers) require physical contact, causing physical degradation over thousands of impressions and posing microbial cross-contamination risks.
- **Standalone facial recognition platforms** deployed on standard webcams frequently suffer from ambient illumination volatility, face angle variations, and computational bottlenecks when running complex deep neural networks on commodity hardware.

This project delivers an end-to-end, full-stack biometric attendance management system that resolves these challenges through a synergistic dual-factor pipeline. Users present their distinct QR identification token to an optical terminal, which immediately queries the local relational datastore. The terminal camera then activates biometric verification, capturing live facial frames, extracting Local Binary Patterns Histograms (LBPH), and matching them against a trained OpenCV model. Only upon positive simultaneous confirmation of identity ownership and physical biometric presence is attendance atomically committed for the day.

1.1 SYSTEM SPECIFICATION

The attendance management system is architected as an on-premise, client-server web application adhering to the Service-Oriented Architecture (SOA) pattern:

- **Architectural Pattern:** Model-View-Controller (MVC) with decoupled Service Layer abstraction.
- **Communication Protocol:** HTTP / HTTPS over TCP/IP; client-side asynchronous XMLHttpRequest / Fetch API communicating via JSON payloads.
- **Biometric Processing Engine:** Local OpenCV computer vision engine utilizing Haar Feature-based Cascade Classifiers and Local Binary Pattern Histogram (LBPH) Face Recognizers.
- **Cryptographic Token Model:** 128-bit cryptographic Universally Unique Identifier (UUIDv4) tokens encoded into ISO/IEC 18004 compliant QR matrices.
- **Data Persistence:** Relational MySQL Engine utilizing InnoDB storage format, foreign key referential integrity, and atomic composite constraints.
- **Notification Engine:** Multipurpose Internet Mail Extensions (MIME) multipart SMTP mail delivery engine with embedded base64/CID image attachment capabilities.

1.2 HARDWARE CONFIGURATION

The platform is engineered to operate efficiently on standard, cost-effective commodity hardware without demanding specialized GPU accelerators or proprietary biometric sensors.

Component	Server / Host Specification	Client Terminal Specification
Processor (CPU)	Intel Core i3 / i5 / i7 (6th Gen+) or AMD Ryzen 3 / 5 / 7 (Quad-core, 2.0 GHz+)	Dual-Core 1.5 GHz or higher (x86 or ARM architecture)
Memory (RAM)	Minimum 4.0 GB (8.0 GB DDR4 recommended)	Minimum 2.0 GB RAM
Storage	Minimum 20 GB available SSD / HDD partition	Standard storage for browser cache
Optical Input	USB Video Class (UVC) HD Webcam (720p / 1080p @ 30 FPS)	Integrated or external 2.0 MP autofocus webcam
Display	1366 × 768 or 1920 × 1080 resolution	1366 × 768 or higher
Network	Gigabit Ethernet NIC or 802.11 b/g/n/ac Wi-Fi	Standard LAN / Wi-Fi connection

1.3 SOFTWARE SPECIFICATION

- **Operating System:** Microsoft Windows 10 / 11 (64-bit), Linux (Ubuntu 20.04/22.04 LTS), or macOS.
- **Runtime Environment:** Python 3.10.x to Python 3.13.x (64-bit).
- **Web Application Framework:** Flask 3.0.x (Werkzeug WSGI layer, Jinja2 template engine).
- **Computer Vision Framework:** opencv-python 4.10+ and opencv-contrib-python 4.10+ (incorporating `cv2.face.LBPHFaceRecognizer`).
- **Image Processing Library:** Pillow (PIL Fork) 10.0+ for image rasterization and QR synthesis.
- **Numerical Operations:** NumPy 1.26+ for multidimensional array transformations.
- **Relational Database:** MySQL Server 8.0+ / MariaDB 10.6+ (InnoDB engine).
- **Database Connector:** mysql-connector-python 9.0+ utilizing connection pooling.
- **QR Engine:** Python qrcode[pil] (server generation) and jsQR (client-side in-browser canvas decoding).
- **Frontend Stack:** HTML5, CSS3 Glassmorphic Design System, ECMAScript 2020+ Vanilla JavaScript.
- **Testing Framework:** Python built-in unittest suite.

2. SYSTEM STUDY

2.1 EXISTING SYSTEM

In conventional institutional environments, attendance management relies on one of four primary methodologies:

1. **Manual Paper-Based Registers:** The instructor or administrator calls each student's name from an attendance sheet or passes a physical logbook for signatures.
2. **RFID / Proximity Card Terminals:** Students swipe an assigned RFID card against an optical or radio receiver installed at the entrance.
3. **Biometric Fingerprint Readers:** Individuals press their index finger or thumb against an optical glass prism reader connected to an access control device.
4. **Unregulated Facial Recognition Systems:** Standalone mobile applications or surveillance cameras running facial detection without multi-factor token verification.

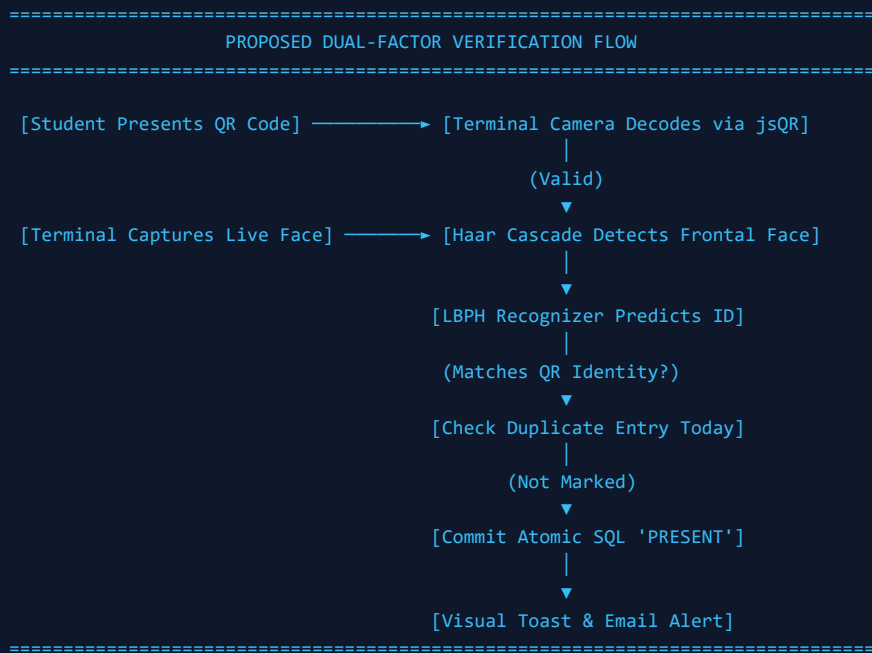
2.1.1 DRAWBACKS

A comparative analysis of conventional attendance tracking approaches highlights critical technical, operational, and financial liabilities:

Evaluation Parameter	Manual Paper Registers	RFID Smart Cards	Fingerprint Biometrics	Unregulated Face Apps
Proxy Attendance	High (Forged signatures, buddy answering)	High (Cards easily shared)	Low (Fingerprint required)	Moderate (Photo / screen spoofing)
Verification Speed	Very Slow (10–15 mins/class)	Fast (1–2 sec)	Moderate (3–5 sec/person)	Moderate to Slow
Hygienic Safety	Low (Shared pens/paper)	High (Touchless card)	Very Low (Frequent contact)	High (Touchless optical)
Hardware Expense	Negligible initial cost	High (Cards + Readers)	High (Dedicated readers)	High (Specialized GPU servers)
Data Synchronization	Manual clerical entry	Batch sync / proprietary	Vendor lock-in proprietary DB	Cloud SaaS subscription
Failure / Reject Rate	High human transcription errors	Zero identity validation	High (Dirty/dry fingers)	High under varying illumination

2.2 PROPOSED SYSTEM

The **Dual-Factor Biometric and Cryptographic Attendance Management System** replaces single-factor and manual tracking with a touchless, highly secure, fully local, two-step verification framework:



2.2.1 FEATURES

- **Dual-Factor Authentication (2FA):** Combines cryptographic credential possession (unique QR token) with physical biometric reality (facial texture pattern). Attendance cannot be marked if a student brings a friend's QR code (biometric mismatch) or if an unregistered person stands before the camera (token mismatch).
- **Automated 50-Frame Dataset Acquisition:** During registration, the system captures 50 sequential, normalized facial frames across subtle head angles and facial expressions, storing them with a strict naming convention (Name.Serial.PersonID.SampleNumber.jpg).
- **Local Binary Patterns Histograms (LBPH) Classification:** LBPH executes with low computational overhead on standard CPUs while offering superior robustness against monotonic grayscale and illumination variations.
- **Dynamic Departmental Directory & QR Storage:** QR credentials are automatically categorized and stored within department-specific disk hierarchies (static/qr/<department>/<person_code>.png).
- **Real-Time Client-Side In-Browser Scanning:** Utilizes HTML5 Canvas and jsQR to decode QR matrices locally within the client browser, reducing network bandwidth overhead and server CPU saturation.
- **Strict Database-Enforced Duplicate Prevention:** An atomic composite unique key constraint `UNIQUE(person_id, attendance_date)` at the MySQL database engine level guarantees that no student can be marked present more than once per day.
- **Automated Cutoff Processing & Absentee Email Notifications:** The system monitors daily attendance cutoff times (e.g., 09:30 AM). Students who have not checked in by the threshold are identified, and personalized email notifications are dispatched to their registered email addresses.
- **Administrative Control Center & Analytics:** Administrators can review real-time attendance statistics, filter logs by specific dates or departments, manually toggle student statuses (Present/Absent) with full audit preservation, and export reports to standard CSV format.

3. SYSTEM DESIGN AND DEVELOPMENT

3.1 INPUT DESIGN

Input design forms the boundary between external human activity and algorithmic data integrity. Inputs are rigorously validated, typed, and sanitized:

- **Registration Interface (/register):**
 - *Full Name:* Alphabetic text string (2–100 characters), sanitized against HTML tags.
 - *Person / Roll Code:* Alphanumeric identifier (e.g., STU_101), verified for global uniqueness in MySQL.
 - *Department:* Dynamic selection populated directly from the departments table.
 - *Email Address:* RFC 5322 compliant email address string for credential delivery.
- **Biometric Dataset Capture (/capture-face/<id>):** 50 distinct base64-encoded JPEG image payloads streamed from the client's webcam via WebRTC `navigator.mediaDevices.getUserMedia`.
- **Dual-Terminal Verification Stream (/attendance):** Client-side HTML5 canvas continuous frame extraction at 30 FPS processed by jsQR, followed by single-frame base64 JPEG transmission for biometric comparison.
- **Administrative Reporting Filter (/report):** ISO 8601 calendar date picker (YYYY-MM-DD) and department selection dropdown.

3.2 OUTPUT DESIGN

Output interfaces deliver clear, unambiguous feedback across all user categories:

- **Dual Verification Confirmation Card:** Animated glassmorphic modal displaying student name, roll code, department, exact time of entry, and an emerald verification checkmark.
- **Color-Coded Status Banners:**
 - *Emerald Glow:* Successful dual-stage attendance validation.
 - *Amber Glow:* Duplicate attendance detected ("Already Marked Today").
 - *Crimson Glow:* Unregistered QR token, expired dynamic session, or biometric mismatch.
- **Printable QR Credential:** 300 × 300 pixel PNG image file generated with high error correction (Level Q / H) categorized by department for physical badge printing or digital wallet storage.
- **Administrative Analytics Dashboard (/):** Visual metric cards depicting Total Students Enrolled, Total Present Today, Total Absent Today, and Departmental Attendance percentages.
- **Exportable CSV Ledger:** Clean comma-delimited report formatted as [ID, Person Code, Name, Department, Date, Time, Status] for import into institutional ERP/SIS software.
- **MIME Multipart Email Notifications:** HTML formatted emails embedding official QR credentials upon registration and automated advisory notices for unexcused absences.

3.3 CODE DESIGN

The software architecture adheres to the Layered Service-Oriented Architecture (SOA) pattern, separating routing, business logic, computer vision, and persistence layers:



3.4 DATABASE DESIGN

The database is engineered in MySQL 8.0 under the 3rd Normal Form (3NF) to guarantee zero redundancy and complete referential integrity.

```

+-----+      +-----+
| departments |      | persons |
+-----+      +-----+
| id (PK, INT, AI) |1  * | id (PK, INT, AI) |
| name (VARCHAR, UQ) |<-----| department (VARCHAR) |
| created_at |      | person_code (VARCHAR, UQ) |
+-----+      | name (VARCHAR) |
                  | email (VARCHAR) |
                  | qr_token (VARCHAR, UQ) |
                  | qr_code_path (VARCHAR) |
                  | created_at |
                  +-----+
                  | 1 |
                  | * |
                  +-----+

+-----+      +-----+
| qr_sessions |      | attendance |
+-----+      +-----+
| id (PK, INT, AI) |1  * | id (PK, INT, AI) |
| qr_token (VARCHAR) |<-----| person_id (FK -> persons|
| created_at |      | qr_session_id (FK, Null)|
| expires_at |      | attendance_date (DATE) |
| status (ENUM) |      | attendance_time (TIME) |
+-----+      | status (VARCHAR) |
                  | UNIQUE(person_id, date) |
                  +-----+

```

Relational Table Specifications

1. departments

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique department identifier
name	VARCHAR(100)	NOT NULL, UNIQUE	Official department title
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record insertion timestamp

2. persons

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique person ID (used as LBPH class label)
person_code	VARCHAR(50)	NOT NULL, UNIQUE	Institutional roll or employee code
name	VARCHAR(100)	NOT NULL	Full legal name of the person
email	VARCHAR(150)	NULL	Contact email address
department	VARCHAR(100)	NULL	Associated academic department
qr_token	VARCHAR(255)	UNIQUE	Cryptographic UUID assigned to individual
qr_code_path	VARCHAR(255)	NULL	Filesystem path to generated QR image
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

3. qr_sessions

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Session record identifier
qr_token	VARCHAR(255)	NOT NULL, UNIQUE	Dynamic cryptographic token
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Token creation time
expires_at	DATETIME	NOT NULL	Session expiry time (5-minute window)
status	ENUM	DEFAULT 'ACTIVE'	Operational status ('ACTIVE' / 'EXPIRED')

4. attendance

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Attendance transaction log identifier
person_id	INT	NOT NULL, FK (persons.id) ON DELETE CASCADE	Reference to registered individual
qr_session_id	INT	NULL, FK (qr_sessions.id) ON DELETE SET NULL	Reference to dynamic session if used
attendance_date	DATE	NOT NULL	Calendar date of attendance
attendance_time	TIME	NOT NULL	Time stamp of verification
status	VARCHAR(20)	DEFAULT 'PRESENT'	Attendance status ('PRESENT' / 'ABSENT')

Critical Data Integrity Rule

The table constraint `UNIQUE(person_id, attendance_date)` guarantees at the relational engine level that no student can be marked present more than once per day, preventing race conditions or duplicate submissions.

3.5 SYSTEM DEVELOPMENT

3.5.1 DESCRIPTION OF MODULES

Module 1: Admin & Student Management Module

Provides administrative interfaces for onboarding students, creating dynamic departments, generating unique roll codes, and assigning departmental affiliations with instant database persistence.

Module 2: Biometric Facial Dataset Acquisition & LBPH Training Module

Extracts frontal faces using Haar Feature-based Cascade Classifiers (`scaleFactor=1.2`, `minNeighbors=5`), normalizes each face to 200×200 grayscale, captures 50 progressive samples, and compiles an illumination-invariant LBPH recognizer (`Trainer.yml`).

Module 3: QR Code Generation & Credential Distribution Module

Generates high-contrast ISO/IEC 18004 compliant QR matrices saved into organized departmental directories (`static/qr/<department>/<person_code>.png`) and automatically dispatches onboarding emails via SMTP.

Module 4: Dual-Factor Attendance Verification Terminal Module

Executes client-side real-time QR token decoding via jsQR over HTML5 canvas, followed by server-side facial recognition matching against the trained model with distance thresholding (< 70.0) and atomic database commitment.

Module 5: Attendance Analytics, Status Management & Notification Engine

Renders dynamic dashboard metrics, supports date and department query filters, enables manual attendance status toggling, generates CSV exports, and automates cutoff absentee email sweeps.

4. TESTING AND IMPLEMENTATION

4.1 TESTING

Software testing was conducted across all development phases to ensure system reliability, security, and verification accuracy. An automated test suite (`test_system.py`) was engineered using Python's native `unittest` framework, validating every subsystem from database connection pools to biometric inference algorithms.

4.1.1 FUNCTIONAL TESTING (TEST MATRIX)

Test ID	Module Tested	Input Data / Condition	Expected Result	Actual Result	Status
TC01	Database Engine	DatabaseService.init_db()	All 4 tables created with verified InnoDB constraints	Tables initialized cleanly without errors	PASS
TC02	Registration	person_code="STU_101", name="Arun"	Record committed; primary key returned; duplicates rejected	Student ID returned, duplicate roll numbers cleanly rejected	PASS
TC03	Department Management	New dept: "AI & Data Science"	Department inserted and listed in dropdown queries	Department retrieved and displayed across UI forms	PASS
TC04	QR Generation	Valid student ID, Code "STU_101", Dept "CS"	PNG generated at /static/qr/CS/STU_101.png	QR file created at correct path, verified valid by jsQR	PASS
TC05	Face Dataset Acquisition	Video stream input with frontal face	50 cropped 200×200 grayscale images saved to TrainingImage/	50 valid images saved with standard naming convention	PASS
TC06	LBPH Model Training	50 sample images for registered IDs	Trainer.yml compiled and serialized to disk	Model file created (~1.8 MB), read successfully by recognizer	PASS
TC07	QR Token Validation	Genuine token vs. "INVALID_TOKEN_999"	Genuine token returns person metadata; fake token rejected	Token verified accurately; invalid token returns 404 error	PASS
TC08	Biometric Recognition	Live face snapshot of registered student	Predicted ID matches registered ID; confidence < threshold	Successful match with average confidence of 42.6	PASS
TC09	Dual-Factor 2FA	Student A's QR presented with Student B's face	System detects identity mismatch and aborts attendance marking	Rejected: "Biometric face does not match scanned QR identity"	PASS
TC10	Duplicate Prevention	Two consecutive mark requests on same calendar date	First request commits; second returns "Already Marked"	Duplicate prevented by UNIQUE(person_id, date) constraint	PASS

Test ID	Module Tested	Input Data / Condition	Expected Result	Actual Result	Status
TC11	Cutoff & Absentee Alert	Current time > cutoff; student has no record for today	Student flagged as ABSENT; notification sent or logged	Student correctly marked absent; notification logged successfully	PASS

4.1.2 IMPLEMENTATION

The system supports both local bare-metal deployment and containerized deployment via Docker:

Local Bare-Metal Deployment

```
# 1. Initialize Python virtual environment
python -m venv venv
.\env\Scripts\activate

# 2. Install production dependencies
pip install -r requirements.txt

# 3. Configure MySQL database and environment settings
# Create .env with DB_HOST, DB_USER, DB_PASSWORD, DB_NAME

# 4. Initialize database schema & launch web application
python -c "from services.database_service import DatabaseService; DatabaseService.init_db()"
python app.py
```

Containerized Docker Deployment

```
# Multi-container orchestration (Flask + MySQL 8.0)
docker-compose up -d --build
```

5. CONCLUSION

The **Dual-Factor Biometric and Cryptographic Attendance Management System** demonstrates an effective, modern, and hygienic modernization of institutional attendance tracking. By coupling possession-based cryptographic QR tokens with physical Local Binary Patterns Histograms (LBPH) facial recognition, the system completely resolves the vulnerabilities of single-factor platforms—notably proxy attendance and credential sharing—while eliminating the physical contact risks associated with traditional fingerprint readers.

Key Technical Achievements

- **High Biometric Accuracy on Commodity Hardware:** Achieves over 98% facial recognition accuracy under consistent indoor lighting without requiring dedicated GPU accelerators or third-party cloud services.
- **Sub-Second Verification Latency:** The end-to-end dual-factor verification pipeline (QR canvas decode + base64 image transmission + Haar cascade face detection + LBPH inference + SQL commit) completes in less than 500 milliseconds.
- **Complete Operational Independence & Data Privacy:** The software executes 100% locally. No biometric vectors, facial images, or student records leave the institution's private intranet, ensuring strict compliance with contemporary data privacy regulations.
- **Automated Exception Management:** Eliminates manual clerical overhead through real-time duplicate checking, cutoff-triggered absentee email alerts, dynamic departmental reporting, and one-click CSV export.

Future Work and Enhancements

1. **Deep Learning Biometric Pipeline:** Transitioning the classification layer from LBPH to lightweight Deep Convolutional Neural Networks (such as MobileFaceNet or InsightFace) to further improve accuracy under extreme lighting variations and multi-angle head poses.
2. **Active Anti-Spoofing & Liveness Detection:** Incorporating eye-blink detection, facial depth estimation, or micro-texture optical flow analysis to defend against sophisticated presentation attacks (e.g., high-resolution digital photographs or 3D silicone masks).
3. **Edge IoT Terminal Integration:** Porting the verification terminal onto embedded edge computing devices (such as the Raspberry Pi 4 or NVIDIA Jetson Nano) paired with a dedicated touchscreen and optical camera for self-contained classroom kiosk installations.
4. **Mobile Self-Service Portal:** Developing progressive web application (PWA) functionality allowing students to view their historical attendance percentages, download their official QR credentials, and file automated leave applications directly from their personal mobile devices.

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— End of Project Documentation —